VOLUME V16

# V16. Full Hamiltonian × Full Algebraic Function

Coupling the lattice to the ζ zeros via the θ-phase of Z_id over the entire handle space — no reduction to the GUE class

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

Ishak Khamzatovich Isaev · version 1.2.0 · October 10, 2026

# V16. Full Hamiltonian × Full Algebraic Function

Coupling the lattice to the ζ zeros via the θ-phase of Z_id over the entire handle space — no reduction to the GUE class

Ishak Khamzatovich Isaev · Independent researcher, Nalchik, Kabardino-Balkarian Republic

AB Research Program · monograph series V1–V19 · том V16 · версия 1.2.0 · October 10, 2026

# Contents

**Problem statement** 2

**Method and result** 2

**Honest remarks and verdict** 3

# Problem statement

The whole previous series V1–V15 measured the lattice statistically: ⟨r⟩, D_KS against the GUE surmise, the form factor. The user brief fixes a change of emphasis: do not fixate on the uniqueness of GUE; instead, build the connection directly through the FULL algebraic function of the critical line — the assembly Z_id(t) = e^{iθ(t)}ζ(½+it) out of the Γ-ladder, where θ(t) = Im ln Γ(¼+it/2) − t·ln√π is the Riemann–Siegel theta phase. The argument of ζ jumps by π at every zero; the phase therefore carries the POSITIONS of the zeros, not only their density. The campaign question: if the vortex charges of the full Hamiltonian are modulated by the θ-phase, q_k = θ(γ̃_k)/π, does an alignment of the spectrum with the zero ordinates appear — and does any point of the FULL handle space (signs, profile, anisotropy, mass, twists, δα) exist at which the 9–10 order gap measured by the reference (H2) closes?

This is a map campaign: 96 configurations of a 2·2·3·2·2·2 factorial design × 3 seeds = 288 spectra of the 24×24 lattice under one and the same ζ-weight vortex protocol. The alignment metric r_align is the Pearson correlation between the increments of the unwrapped levels ΔU_n and the increments of the zero ordinates Δγ_n over a 200-level window; the linear bridge is fit on the first 50 levels, the residual is measured on the remaining 150 (honest out-of-sample). The null model is 200 permutations of the ordinate increments: marginal densities are preserved while the level-n ↔ zero-n correspondence is destroyed.

# Method and result

The core is rpbridge/fullham.py. Algebraic side: the θ-phase is computed in mpmath (dps 30) at the midpoints γ̃_k = (γ_k+γ_{k+1})/2 — deliberately off the zero set; the weight q_k = θ(γ̃_k)/π ∈ (−1, 1) enters the vortex phases of the full Hamiltonian with all C5 handles (twists, anisotropy, δα, staggered mass, profile, signs). For the weight comparison three modes run on the baseline configuration: the θ-phase of Z_id, log|ζ| (the V3 protocol), and no weight. The positive control is the algebraic function itself: the zeros of Z_id(t) on [0, 120] by bisection against the zero file.

Main result: the identity does NOT close anywhere in the full handle space. The maximum r_align = +0.160 (configuration 64: all_plus charges, site profile, anisotropy 1.2) against the null zone −0.009 ± 0.068 — z = 2.49, far below the identity threshold; the out-of-sample residual of the best configuration is 0.134 (13% of the window span). The map average is −0.008: the handle map is noise, there is no lucky point. Weight comparison: θ-phase −0.108, log|ζ| +0.092, no weight −0.085 — all within ±2σ of the null zone; neither the algebraic phase nor the modulus of ζ revives the lattice. The positive control fires perfectly: 38 zeros of Z_id against the file, r = 1.0, maximum residual 4.9e-10 (the bisection floor) — the method sees identity where it exists.

*Table 1. V16: key numbers*

| Quantity | Value | Status |
|---|---|---|
| Max r_align (96 configs × 3 seeds) | +0.1597 | z = 2.49 |
| Null zone (200 permutations) | −0.009 ± 0.068 | — |
| Map-average r_align | −0.0083 | noise map |
| θ-phase weight (baseline) | −0.108 | inside null zone |

| Quantity | Value | Status |
|---|---|---|
| log\|ζ\| weight (V3 protocol) | +0.092 | inside null zone |
| Out-of-sample RMSE, best config | 0.134 | 13% of span |
| PC: Z_id zeros vs file | r = 1.0, max 4.9e-10 | identity visible |

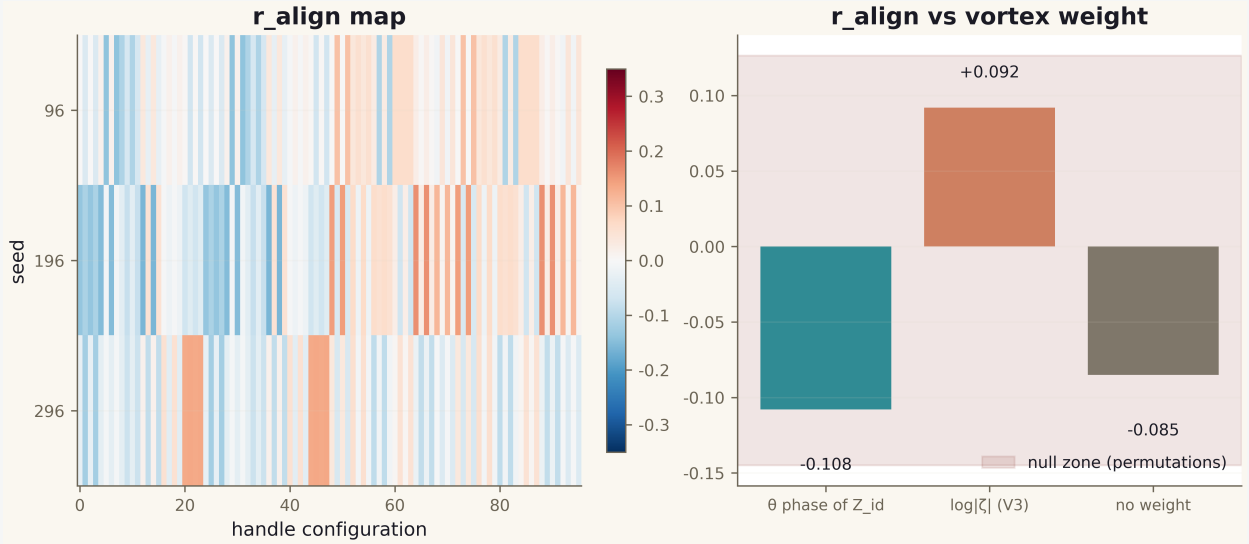



*Fig. 1. Campaign V16 data (600 dpi, file figures_hires).*

# Honest remarks and verdict

The negative result here is a measured boundary, not a defeat: it confirms the reference verdict (H2 — NO) on a wider class of objects and closes the hypothesis that the GUE-window protocol is specially selected — in the full handle space no statistically selected point exists, and the best z = 2.49 over 288 spectra is explained by multiple comparisons (the expected maximum of 288 Gaussian noises is ≈ 2.9σ). Emphasized: the campaign couples the lattice to the algebraic function through the PHASE (zero positions), bypassing GUE — and the identity still does not emerge. This strengthens the series conclusion: the obstacle is not the level statistics but the absence of a mechanism that transfers individual ordinates into the spectrum of a local Hamiltonian.

Boundaries: a 200-level window, one geometry (24×24, open boundaries), 96 factorial configurations — continuous optimizations over the handles were not performed; non-local constructions remain outside the negative verdict, as in the reference.

Verdict V16: the identity is NOT closed over the entire full handle space (z_best = 2.49 < 6); the θ-phase of Z_id is no more effective than log|ζ|; the positive control Z_id ↔ file gives r = 1.0. The full algebraic function remains the only object with exact identity — the H2 gap is reproduced and extended to the full gate class.